

Power System Analysis

EE2600

Exam 4

Date: 02-05-2023

Answer all the questions. If required, make suitable assumptions and state those. Please try to maintain computational accuracy upto four decimal places.

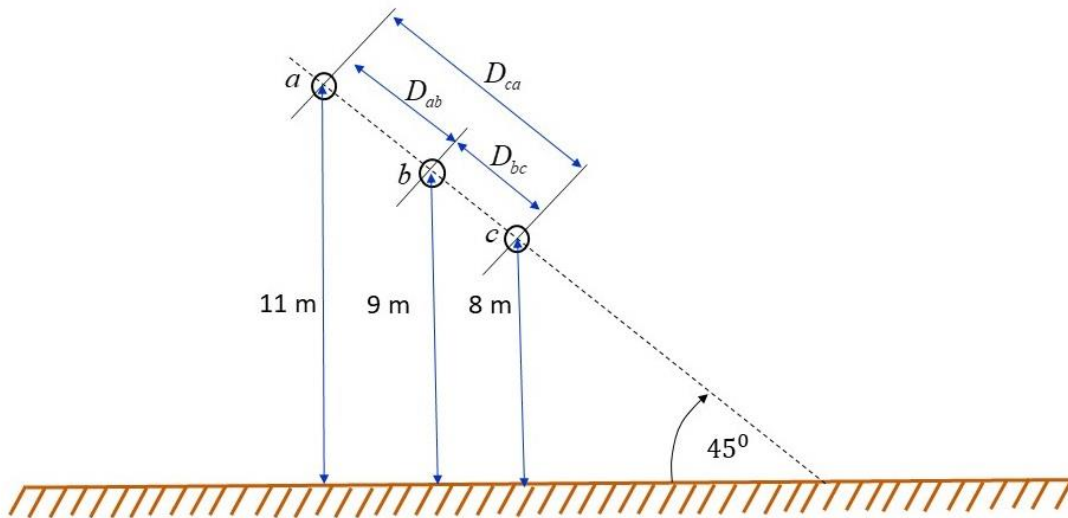
Time: 3 hours

Full marks: 30

Q1.

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Each phase of a transposed three-phase transmission line comprises only one solid conductor. The radius of each conductor is 2 cm. The arrangement of conductors over a particular transposition section is shown below. Find out the GMD between two conductors over a complete transposition cycle as well as the GMR of a conductor. Subsequently find out the effective self-inductance of a phase and the effective mutual inductance between two phases per unit length of the transmission line. **Please provide the answers for inductances in $\mu\text{H}/\text{m}$.**



Q2.

5

Consider a 500 km long 50 Hz lossless distributed parameter transmission line of series inductance equal to 1.4 mH/km and shunt capacitance equal to 9.51 nF/km. The current carrying capacity of the transmission line is 0.5 kA. Evaluate the voltage regulation performance (in percentage) of the transmission line at 0.8 (lagging) load power factor (i.e., power factor at the receiving end) if the full-load receiving end voltage magnitude at the corresponding load power factor is 170 kV. Note that, here, the full-load operation means that the receiving end current is at the maximum limit. It is also to be noted that the sending end voltage magnitude should be a fixed quantity while evaluating the voltage regulation performance of a transmission line.

Q3.

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Draw a phasor diagram showing the Ferranti effect in a distributed parameter transmission line for an inductive load at the receiving end. Subsequently, draw another phasor diagram demonstrating the Ferranti effect for the capacitive load.

Q4.

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Refer to the optimization problem shown below. It is guessed that the first inequality constraint should be inactive and the second inequality constraint should be active at the optimal solution point. Find out the solution for primal and dual variables on the basis of the above assumption. You also need to verify whether the solution thus obtained is really the optimal solution or not. The whole analysis should be carried out on the basis of the KKT necessary conditions of optimality. Please do not forget to separately write down the Lagrangian function and the KKT necessary conditions optimality for the given problem. Please note that no sufficiency condition verification is required here because this is a convex optimization problem.

$$\begin{aligned} & \text{Minimize } \{3x^2 + 2y^2 - 4xy - 1.5x - 10y\} \\ & \text{s.t.} \\ & \quad 3x - 2y \leq 8 \\ & \quad 14 - x - 5y \geq 0 \end{aligned}$$

Q5.

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Use the Newton-Raphson method to solve the following simultaneous non-linear algebraic equations. Report the results of first two iterations. The initial values of x and y should be taken as 1.5 and 0.5, respectively. No acceleration factor should be used. Do not forget to show the expression of the Jacobian matrix.

$$\begin{aligned} x^2y + xe^y &= 2 \\ x^2 + y^2 &= 4 \end{aligned}$$

Q6.

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Derive an expression of the reactive power loss in a transmission line. The expression should be derived in terms of the terminal bus voltage magnitudes and angles by using the equivalent pi-model parameters. You can simply use the medium-length transmission line model. For the sake of simplicity, the shunt conductance term can be ignored.

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